ENTERPRISE FOUNDATIONS

- Introduction
- Enterprise software characteristics
- Component Based software development for enterprise
- Enterprise Architectural overview
- Multi-tier system
- Java Enterprise System
- Use of patterns, frameworks
- Software stacks for Enterprise application development

Enterprise Computing

- Introduction
- Components of an enterprise
- Enterprise Categories
- Characteristics of an Enterprise Computing
- Component based system
- Overview of an enterprise architecture
- Challenges faced by Enterprise Applications
- Technology – outline (J2EE, .NET Frame work, Software Stack)

ENTERPRISE

Enterprise- A business organization

In Computer Industry, the term is often used to describe any large organization that utilizes computers

What is Enterprise Computing?

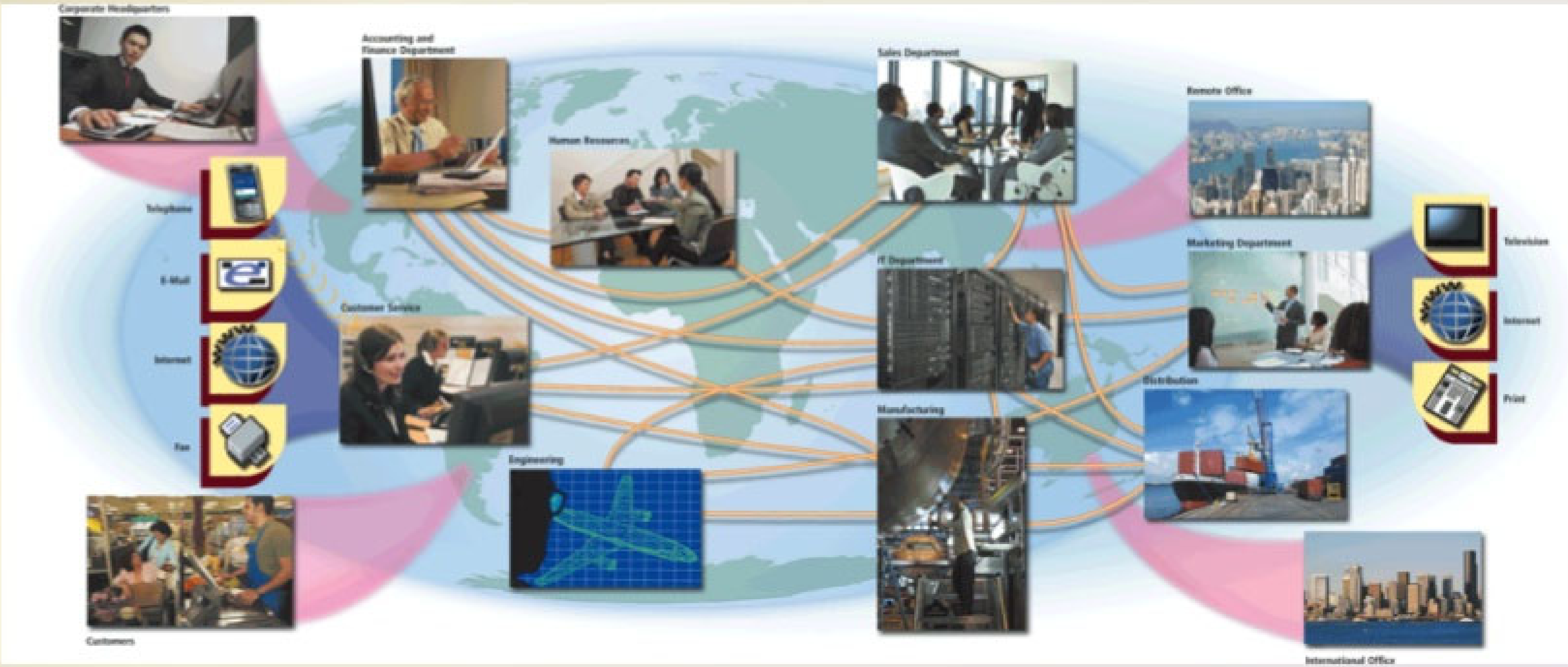
Large organization such as multinational corporation, university, hospital, research laboratory, or government organization

Requires special computing solutions because of its size

Enterprise computing—use of computers in networks that encompass variety of operating systems, protocols, and network architectures

What Is Enterprise Computing?

- **Enterprise computing** involves the use of computers in networks such as LANs and WANs, or a series of interconnected networks that encompass a variety of different operating systems, protoco and network architectures



What Is Enterprise Computing?

- Types of enterprises include:

Retail

Manufacturing

Service

Wholesale

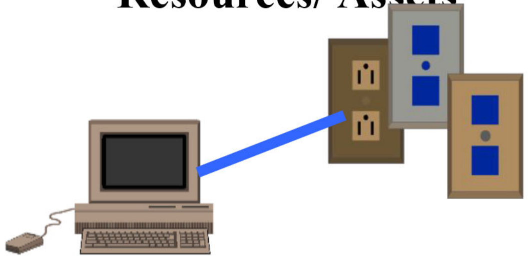
Government

Educational

Transportation

Components of an Enterprise

Resources/ Assets

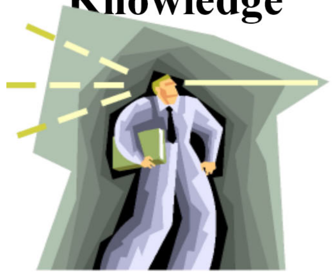


People/Users Partners
Employees/Contractors

Information

Customers

Knowledge



Enterprise Categories

- **Size**

  Small, Medium, Large

- **Profit** - Commercial / Non-profit/ Government

- **Domain**

  Manufacturing, Financial, Health and Education

- **Scope**

  Local, National , MNC

- **Lifetime**

  Startup, Corporate

Characteristics of an Enterprise Computing

- Complex System
- Scalable system
- Distributed Computing
- Fault tolerance
- High availability
- Component based system

Complex System

- A **complex system** is a system composed of many components which may interact with each other.
  - New functionalities are added to existing components
  - New components are created and must be integrated with existing ones
  - New technologies are introduced
  - Old technologies are replaced

❖ Quality of service has to be maintained

Scalable System

- Scalability **is** a characteristic of an organization, system, model, or function that describes its capability to cope and perform well under an increased or expanding workload or scope.

Distributed Computing

Distributed computing is a model in which components of a software system are shared among multiple computers.

- Even though the components are spread out across multiple computers, they are run as one system.
- This is done in order to improve efficiency and performance.
- A distributed system can provide more reliability than a non-distributed system, as there is no single point of failure.

Fault tolerance

- Fault tolerance refers to the ability of a system to continue operating without interruption when one or more of its components fail.
- The objective of creating a fault-tolerant system is to prevent disruptions arising from a single point of failure, ensuring the high availability and business continuity of mission-critical applications or systems.

High availability

- High availability refers to a system’s ability to avoid loss of service by minimizing downtime.
- In most cases, a business continuity strategy will include both high availability and fault tolerance to ensure your organization maintains essential functions during minor failures, and in the event of a disaster.

Component based system

A component is a modular, portable, replaceable, and reusable set of well-defined functionality that encapsulates its implementation and exporting it as a higher-level interface.

Component frameworks such as JavaBean, EJB, CORBA, .NET and web services

- **Reusability** − Components are usually designed to be reused in different situations in different applications.
- **Replaceable** − Components may be freely substituted with other similar components.
- **Not context specific** − Components are designed to operate in different environments and contexts.
- **Extensible** − A component can be extended from existing components to provide new behavior.

- **Independent** – Components are designed to have minimal

What is enterprise architecture?

- **Enterprise architecture is the process by which organizations standardize and organize IT infrastructure to aligns with business goals.**
- **Any Business Innovation and Transformation requires an Enterprise Architecture**
- Everyone realizes that organizations have to be innovating or continually transforming themselves to be competitive in a changing world like the present

CHALLENGES FACED BY ENTERPRISE APPLICATIONS

- Business Process Automation
  - -Automating business process
- Data Harmonization
  - -Unification and standardization of data
- Application Integration
  - -Integrating individual application
- Application Security
  - -build up the security perimeter
- Internationalization
  - -Handling Globalization in localized manner

CHALLENGES FACED BY ENTERPRISE APPLICATIONS

- Transaction Management

- Managing transactions from customers, vendor partners, suppliers and stakeholder.

- Rich User Experience

  - Users become more demanding and experienced.

- - Quality of Service (QoS)
- Technology Selection

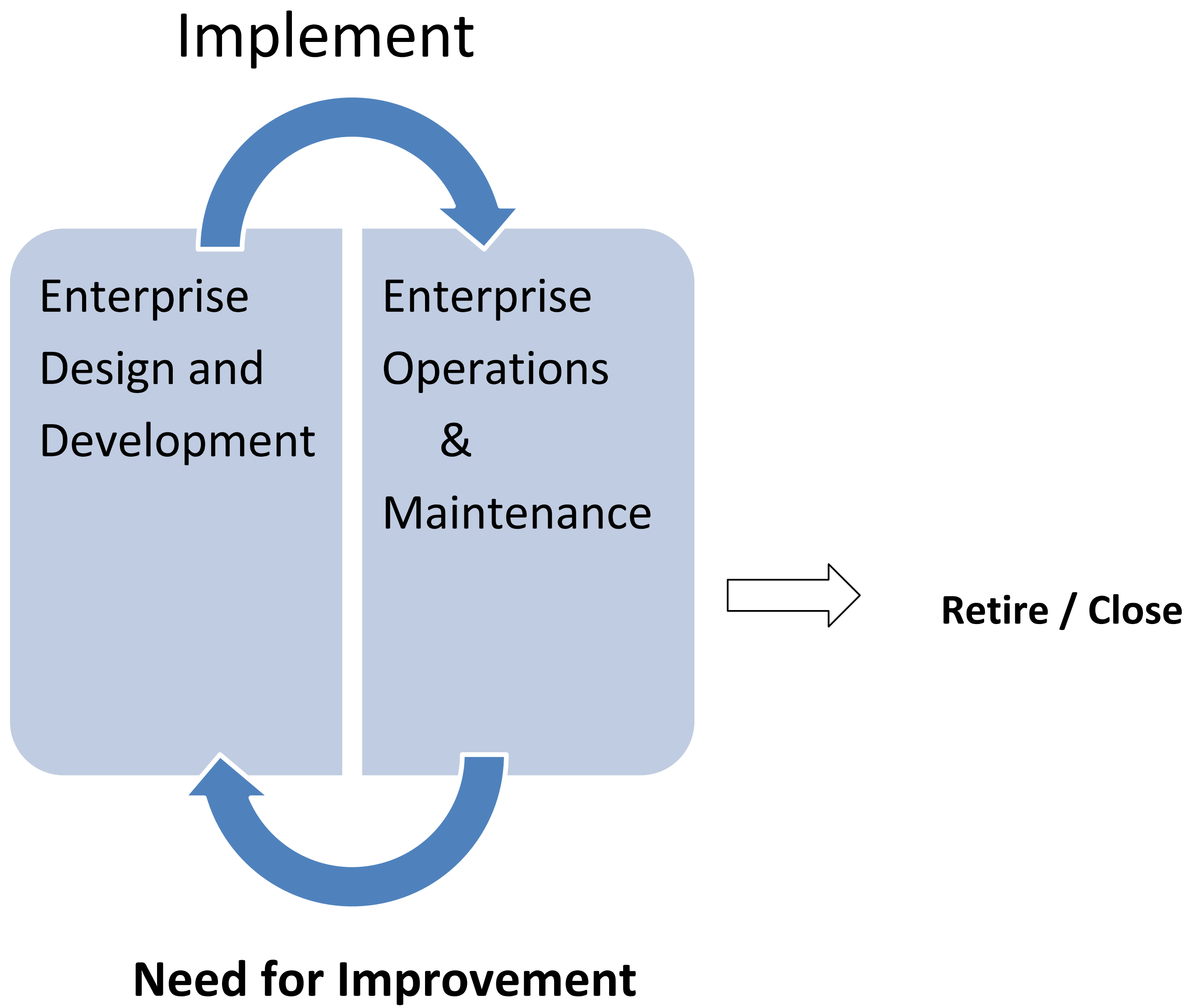
  choosing right technology

- Functional and non-functional quality attributes like scalability, reliability, security, maintainability, availability etc.

Enterprise Life Cycle (ELC)

Dynamic, iterative process of changing the enterprise over time by incorporating new business processes, new technology, maintenance, Disposition and disposal of existing elements of the enterprise.

Enterprise Life cycle



Factors affecting the survival of an Enterprise

- Endogenous Reasons (INSIDE factors)
  - Inefficient Operations
  - Higher Internal Costs
  - Lack of Coordination among various stakeholders
  - Poor Customer Services
  - Missed opportunities
- Exogenous Reasons (OUTSIDE factors)
  - Globalizations
  - Extreme Competitions
  - Consumerization
  - Rapidly Changing Technology
  - Forces beyond our Control
  - Natural Disaster like Pandemic

Enterprise-Wide Technologies

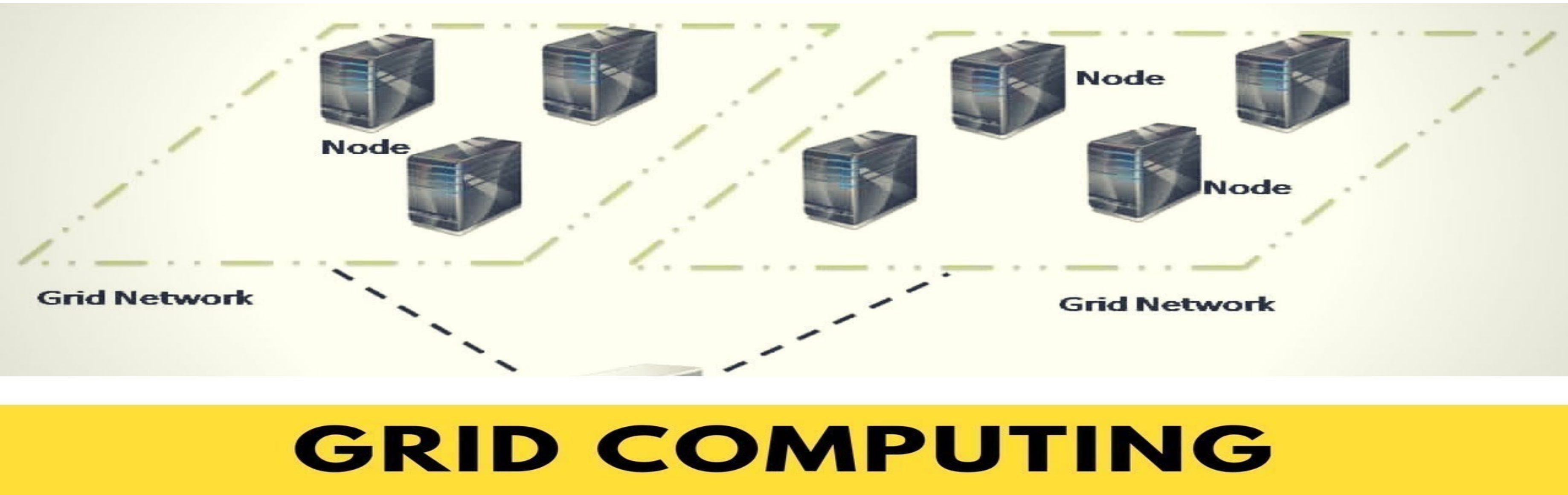
- **Grid computing**
- **Cloud computing**
- **Blade servers**
- **Thin clients**
- **Web portals**
- **Electronic data interchange**
- **Virtual private networks**
- **Intranets and extranets**
- **Computer-based and Web-based training**
- **Teleconferencing**
- **Telecommuting**
- **Workgroup computing**

Enterprise-Wide Technologies

- **Intranet**
  - Internal enterprise network
  - Accessed only by employees or authorized individuals
  - Employee manuals and telephone directories can be moved to an intranet, creating enormous savings for the enterprise
- **Extranet**
  - Network that enables outside users to access an intranet through the Internet
  - Data transfer is encrypted

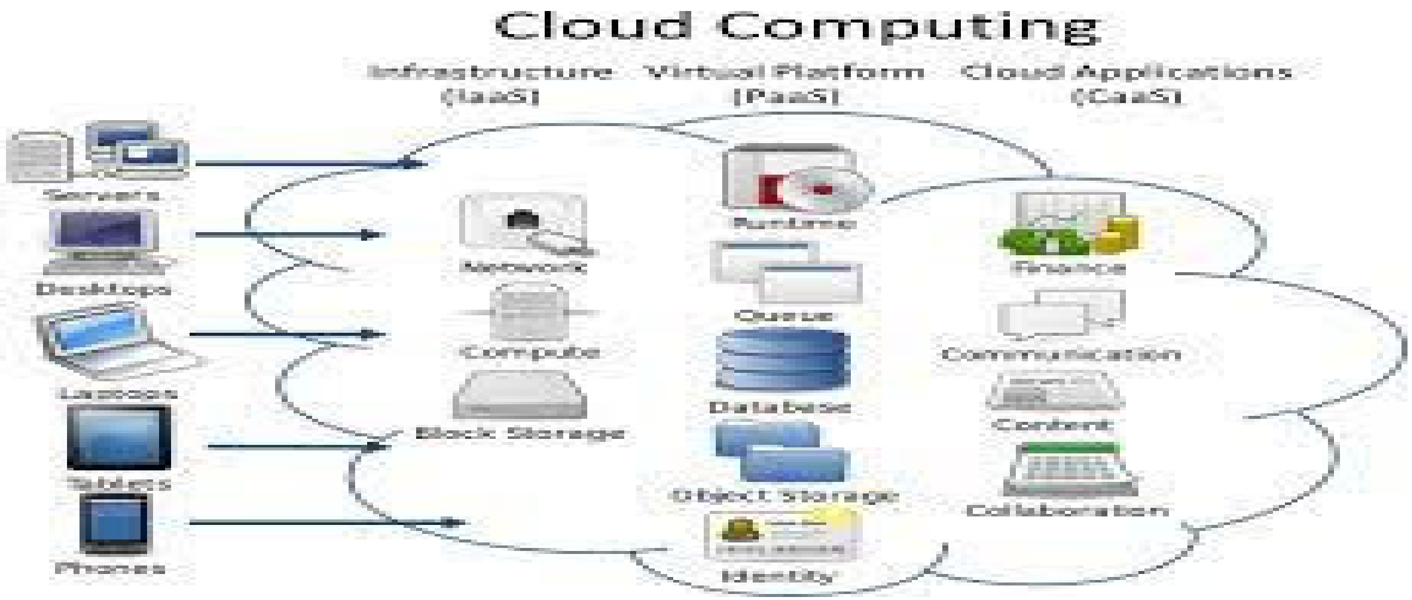
Grid computing

- Network of computers working together to perform a task that would rather be difficult for a single machine.



Cloud computing

- on-demand availability of computer system resources, especially data storage (cloud storage) and computing power, without direct active management by the user.



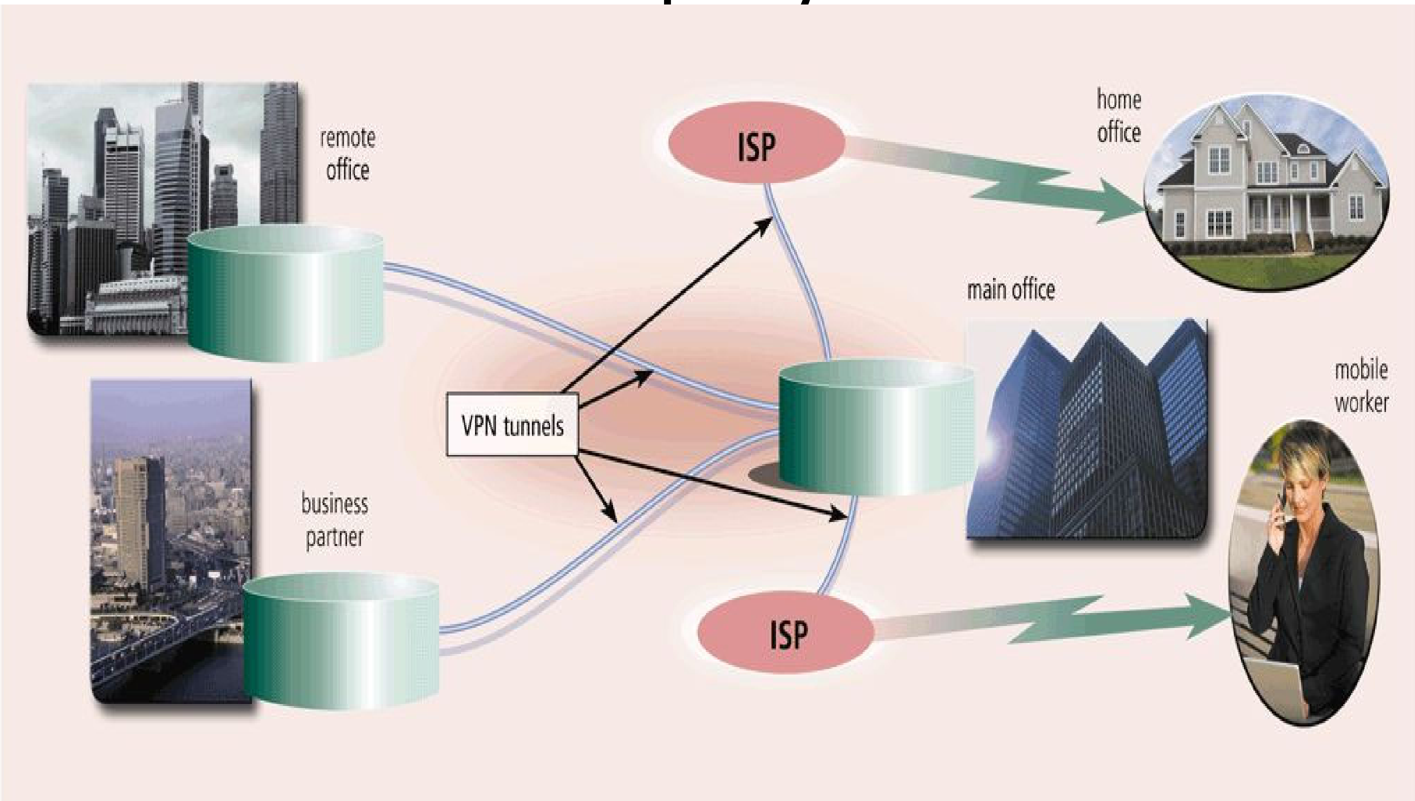
Virtualization and Cloud Computing

Cloud computing is an Internet service that provides computing needs to computer users

Grid computing combines many servers and/or personal computers on a network to act as one large computer

Enterprise-Wide Technologies

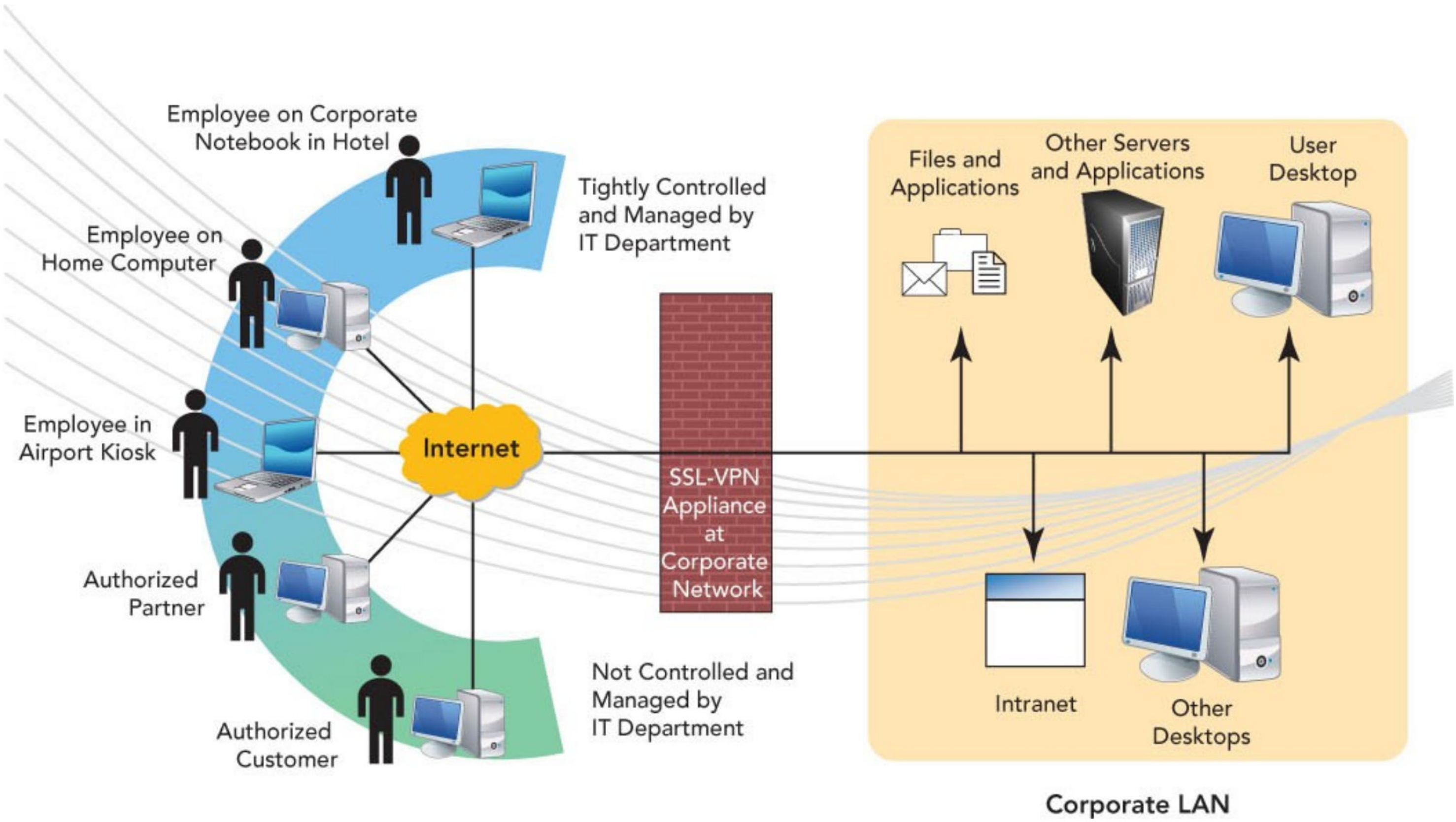
- A **virtual private network** (**VPN**) provides mobile users, vendors, and customers with a secure connection to the company network server



Enterprise-Wide Technologies

Enterprise-Wide Technologies

- **VPNs**



Enterprise-Wide Technologies

- What is an extranet?

Portion of network that allows customers or suppliers to access parts of enterprise's intranet

EDI (electronic data interchange)—Set of standards that controls transfer of business data among computers

Enterprise-Wide Technologies

EDI is a set of standards that controls the transfer of business data and information among computers both within and among enterprises

An **extranet** is the portion of a company's network that allows customers or suppliers of a company to access parts of an enterprise's intranet

Virtualization and Cloud Computing

- Virtualization is the practice of sharing or pooling computing resources

Server virtualization

- Provides the capability to divide a physical server logically into many virtual servers

Storage virtualization

- Provides the capability to create a single logical storage device from many physical storage devices

Enterprise-Wide Technologies

- **Blade servers**
  - Modular server that allows multiple servers to be housed in a smaller area
  - Energy efficient low-cost modular computers

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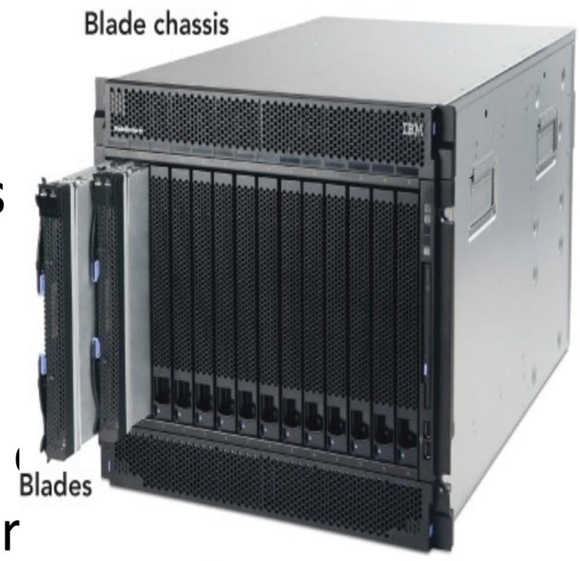
- **Thin client**
  - Software program or computer that relies on other computers

to do most of the work

  - The server does most of the work, which include launching software programs, per calculations, and storing data.



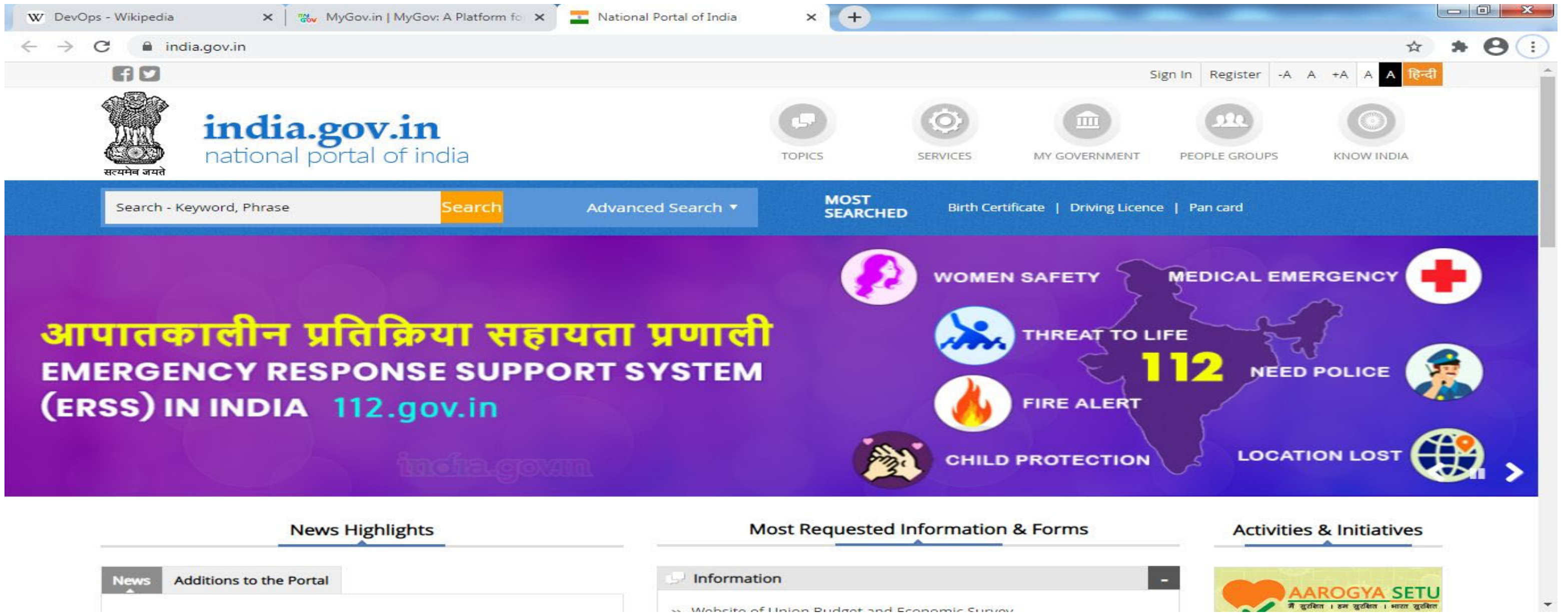
Enterprise-Wide Technologies

Enterprise-Wide Technologies

- **Web portals**
  - Web sites that supply numerous online services
  - Examples: AOL, Yahoo!, MSN, and Google
  - Business portals offer centralized knowledge and content management

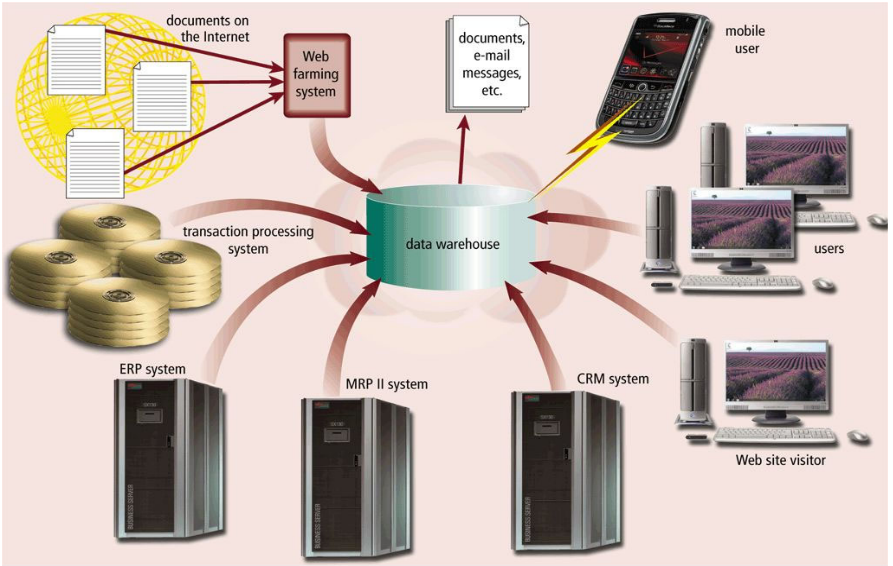

documents
drawings
multimedia
data
SanDisk
SDHC Card
8GB
information from other internal sources
process
categorize
index
store
information from external sources such as news and journals
enterprise portal software
financial reports
legal documents
e-mail, IM, and text messages
resumes and job requests
financial information
orders
print
fax
wireless
systems
HR
sales
executives

PORTAL- Enterprise-Wide Technologies

- A *web*-based platform that collects information from different sources into a single user interface and presents users with the most relevant information
- 

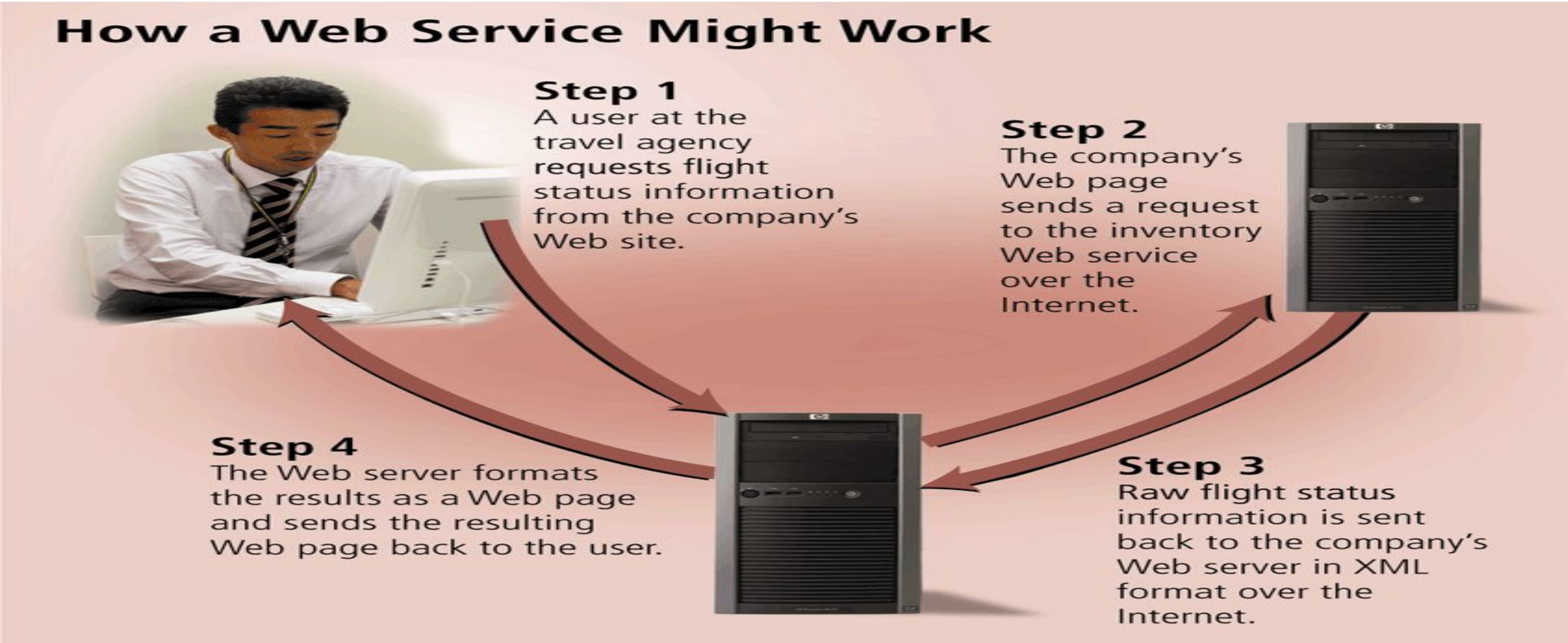
Enterprise-Wide Technologies

- A **data warehouse** is a huge database that stores and manages the data required to analyze historical and current transactions



Enterprise-Wide Technologies

- **Web services** allow businesses to create products and B2B interactions over the Internet



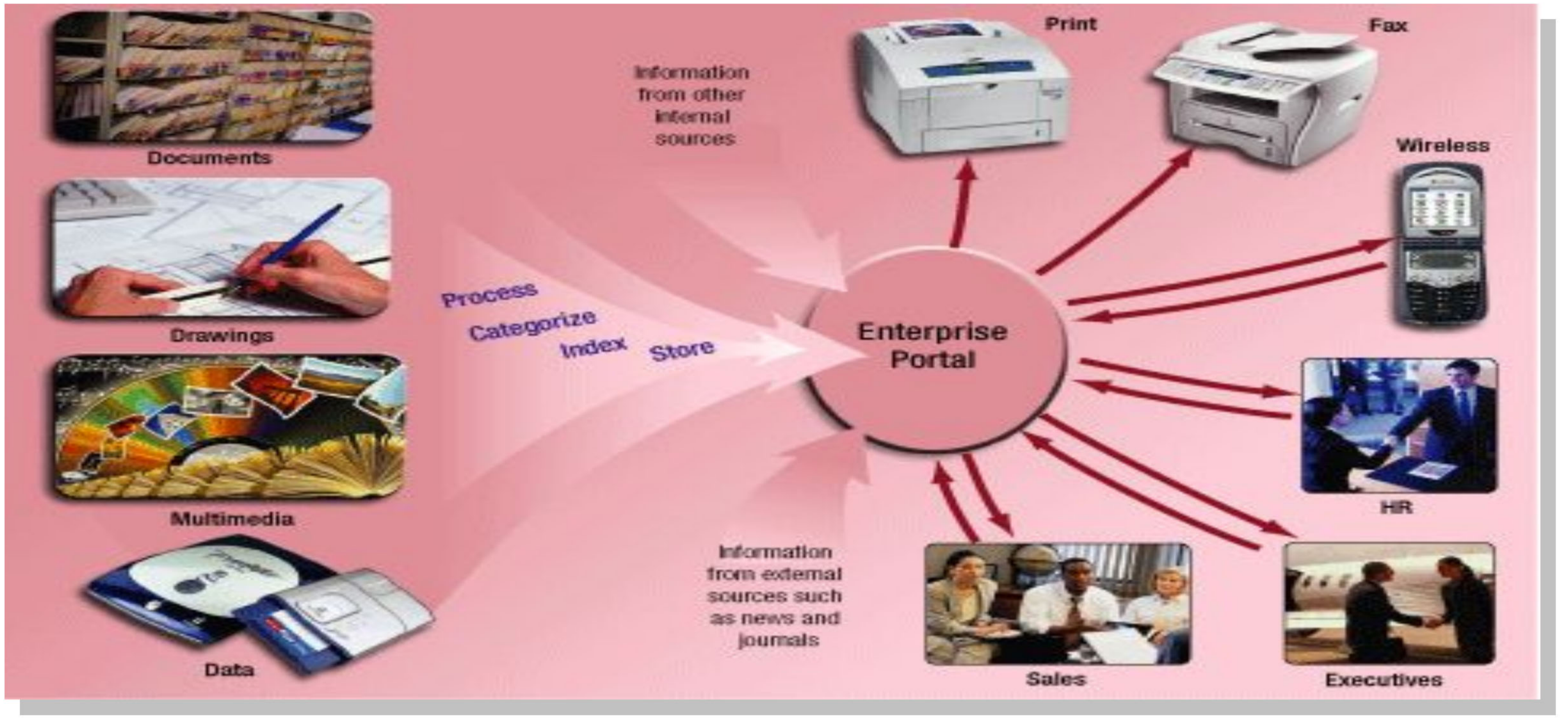
Enterprise-Wide Technologies

- A **document management system** (**DMS**) allows for storage and management of a company's documents
  - Stored in a repository
- In a service-oriented architecture, information systems provide services to other information systems in a well-defined manner over a network

Information Systems in the Enterprise

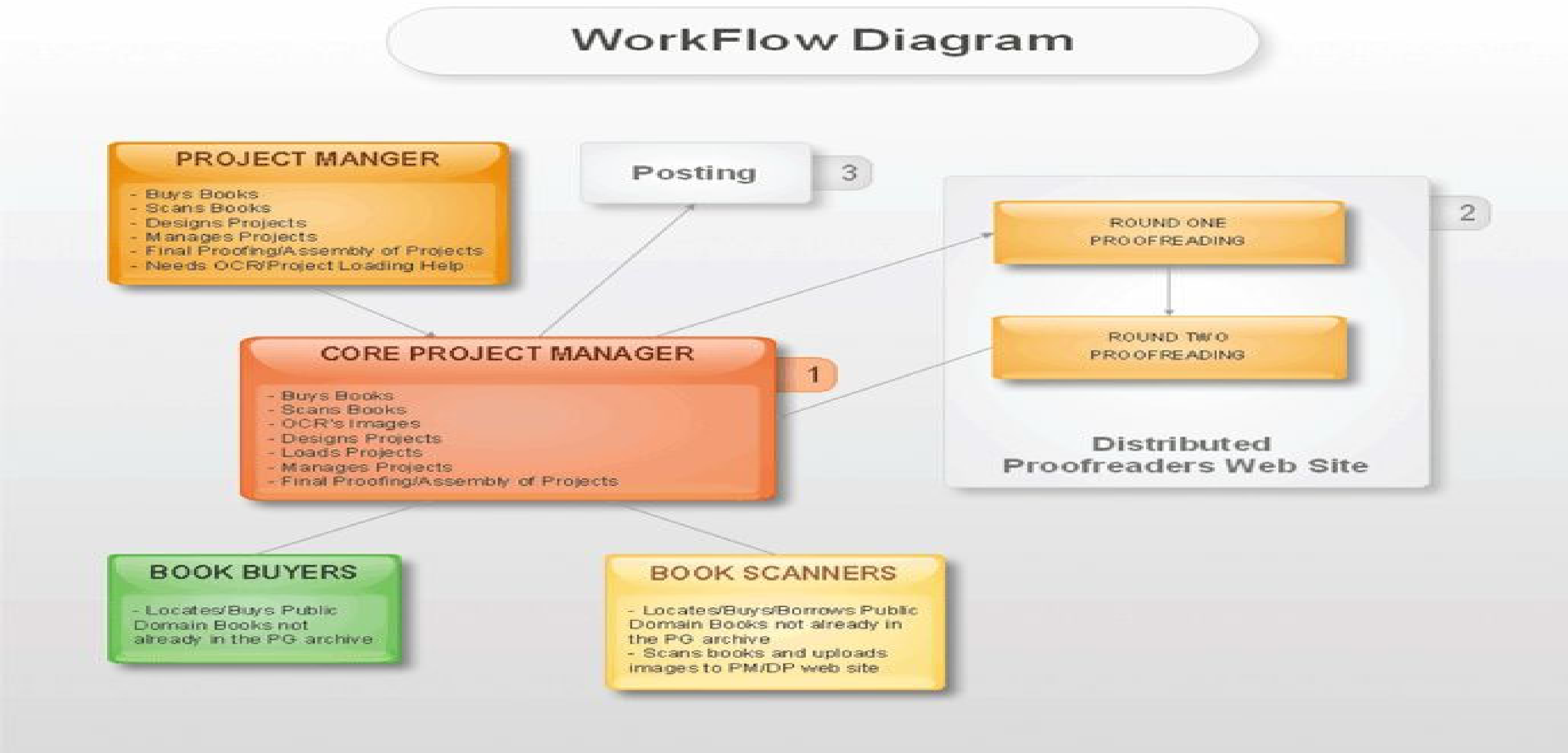
content management system (CMS)

- Combination of databases, software, and procedures
- Organizes and allows access to documents and other files



Enterprise-Wide Technologies

- A **workflow** is a defined process that identifies the specific set of steps involved in completing a particular project or business process



Enterprise-Wide Technologies

Workflow

- **Steps involved in completing project or business process**
- **Workflow application is a program that tracks process from start to finish**

Enterprise-Wide Technologies

- **Telecommuting**
  - Working from home via computer and telecommunications equipment
  - Numbers are increasing
  - Benefits—productivity gains, lower employee turnover, and reduced costs for office space
    - Disadvantage—lack of direct supervision of employee

Enterprise-Wide Technologies

- **CBT (Computer-based training)**
  - Convenient, affordable, learning method
  - Useful when multimedia, animation, and programmed learning are used
- **WBT (Web-based training)**
  - Similar to CBT
  - Training provided over the Internet or intranet
    - Includes instant messaging, discussion forums, chat tools, Web broadcasts with streaming audio or video, and videoconferencing

Enterprise-Wide Technologies

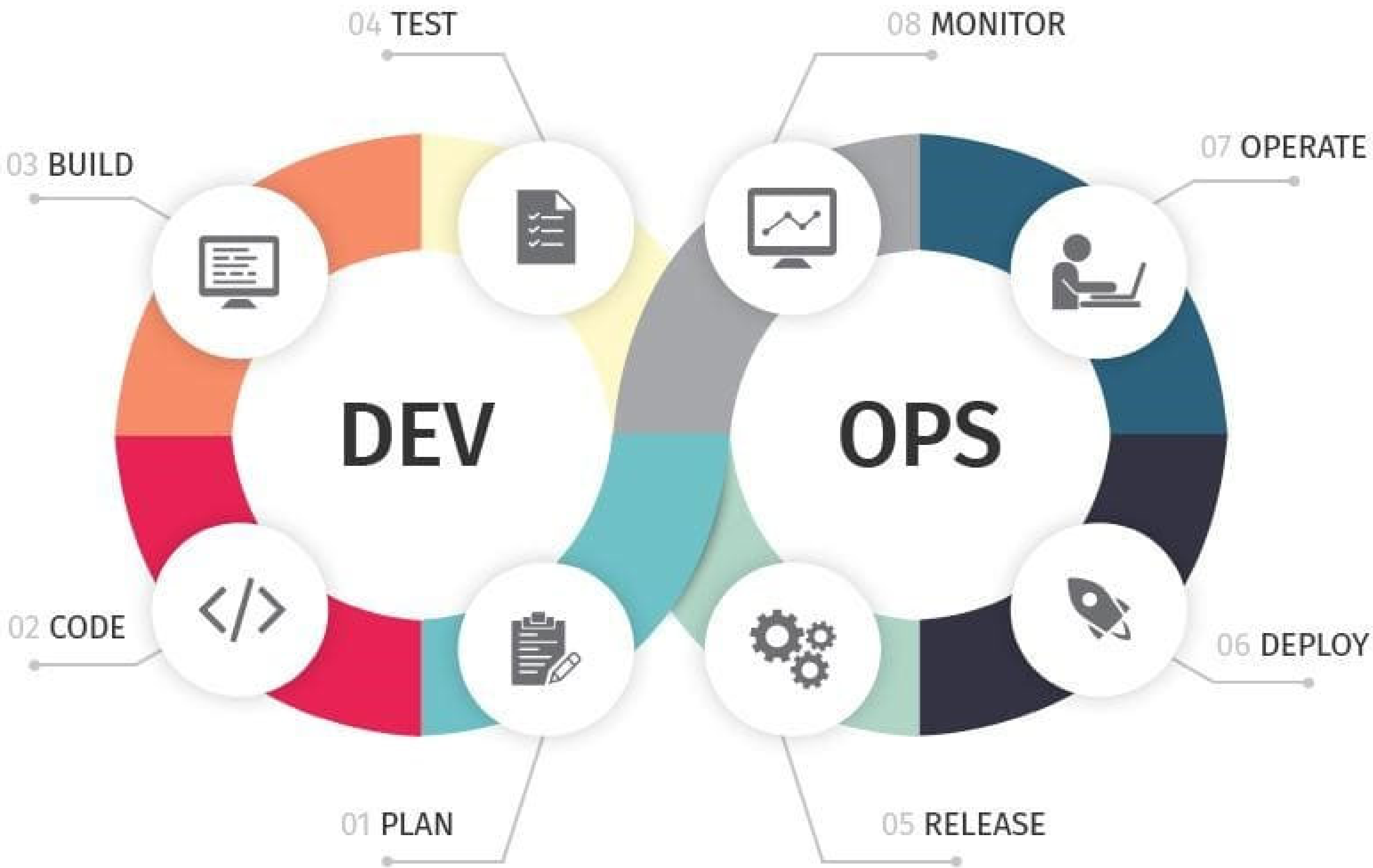
- **Teleconferencing**
  - Enables business to be conducted by using computer and telecommunications equipment
  - Enhances enterprise communication
  - Often reduces costs

Enterprise-Wide Technologies

- **Workgroup computing**
  - Used to communicate and collaborate
  - Uses specific computer hardware, software, and network equipment
  - **Groupware (teamware)** is software that supports workgroup information requirements.
    - Applications include:
      - E-mail
      - Videoconferencing tools
      - Group-scheduling systems
      - Customizable electronic forms
      - Real-time shared applications
      - Shared information databases
  - Facilitates **workflow automation**

DevOps

- **DevOps** is a set of practices that combines software development (*Dev*) and IT operations (*Ops*). It aims to shorten the systems development life cycle and provide continuous delivery with high software quality.
- A compound of development (Dev) and operations (Ops),
- DevOps is the union of people, process, and technology to continually provide value to customers.


04 TEST
08 MONITOR
03 BUILD
07 OPERATE
DEV
OPS
02 CODE
06 DEPLOY
01 PLAN
05 RELEASE

Confluence
JIRA
git
docker
aws
CHEF
ANSIBLE
kubernetes
sbt
Maven
JUnit
Jenkins
Nagios
splunk
DATADOG
CODE
BUILD
TEST
RELEASE
DEPLOY
OPERATE
MONITOR

Technology - outline

- The **Java EE** (J2EE) stands for **Java Enterprise Edition**, which is a standalone **Java** environment used by the software developers for building and deploying a web-based application or a website. J2EE technology provides a platform for developers with enterprise features such as distributed computing and web services.

.Net Framework

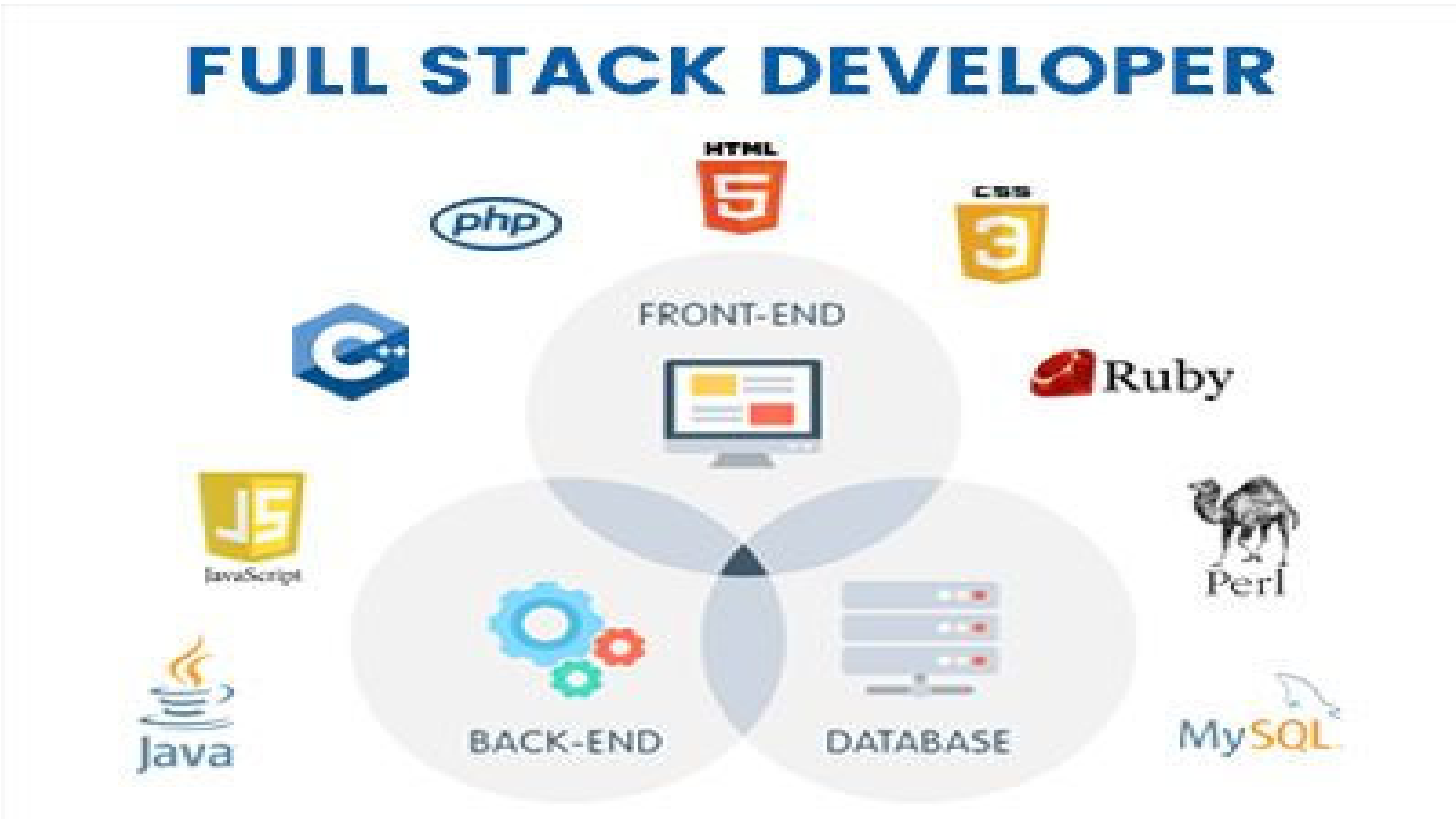
- **.Net Framework** is a software development platform developed by Microsoft for building and running Windows applications. The .Net framework consists of developer tools, programming languages, and libraries to build desktop and web applications. It is also used to build website and web services.

software stack

- A **software stack** is a collection of independent components that work together to support the execution of an application. The components, which may include an operating system, architectural layers, protocols, runtime environments, databases and function calls, are stacked one on top of each other in a hierarchy.

MEAN - Stack

- It is an open-source, JavaScript software stack used for building dynamic enterprise apps and websites.
- The MEAN stack consists of MongoDB, Express.js, Angular JS (or Angular), and Node.js. In MEAN stack, development of both server-side and client-side applications have language uniformity as the support programs are coded in JavaScript.
  The variation of MEAN includes MEEN - MongoDB, Express.js, Ember.js, and Node.js.
- MERN stands for MongoDB, Express, React, Node, after the four key technologies that make up the stack.


FULL STACK DEVELOPER
HTML
CSS
php
FRONT-END
Ruby
JavaScript
Perl
BACK-END
DATABASE
Java
MySQL

Few List of Enterprise Solution Providers

ENTERPRISE SOFTWARE CATEGORY	EXAMPLE SOLUTION	CAPTERRA RATING
Business Intelligence	datapine	4.7 *****
Customer Relationship Management	Salesforce	4.4 ****
User Feedback	Mopinion	4.5 *****
Online Payment	Stripe	4.7 *****
HR Recruiting	iCIMS	4.3 ****
Ideation	IdeaScale	4.6 *****
Instant Messaging	Slack	4.6 *****
Online Marketing	Moz	4.4 ****
Project Management	Monday	4.6 *****
Web Analytics	Google Analytics	4.7 *****
Enterprise Resource Planning	Sage	4.4 ****
Customer Service	Zendesk	4.4 ****
Issue Tracking	Jira	4.4 ****
Surveys	SurveyMonkey	4.6 *****
Cloud Computing	AWS	4.7 *****